Flooding and Tourism in Venice, Italy: Evaluating the Economic Justification for the MOSE Barrier

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I. Introduction

Venice, Italy – one of the world's most iconic tourist destinations and a city of cultural and architectural significance – has long stood as a symbol of the tension between historic preservation and environmental vulnerability. Designated a UNESCO World Heritage Site in 1987, Venice is recognized not only for its iconic canals, Renaissance architecture, and historic urban fabric, but also for the fragility of the lagoon ecosystem on which it was built (UNESCO 2014). In recent years, UNESCO has warned that Venice is at increasing risk of "irreversible damage" due to climate change, rising sea levels, and unsustainable tourism (UNESCO 2023).

A central environmental problem facing Venice is the increasingly frequent phenomenon of "*acqua alta*", or high water (City of Venice 2022). In the Venetian context, flooding officially begins when tidal levels exceed 110 centimeters above average sea level, submerging approximately 14% of the city. As tides rise further, to 120, 130, or 140 cm, the share of the city affected increases sharply, with 130 cm flooding over 70% of urban space (City of Venice 2011).

These events are not merely environmental concerns; they disrupt transportation, damage infrastructure, and render large portions of the city inaccessible. When floods occur, especially during peak travel months, tourists may cancel visits, shorten stays, or avoid the city entirely, leading to significant losses in revenue for hotels, restaurants, cultural sites, and local businesses.

Given that Venice recorded approximately 5.7 million overnight tourist arrivals in 2023, and that tourism accounts for a significant portion of its local economy, the economic consequences of flooding are substantial and policy-relevant (City of Venice 2023). As climate change continues to raise global sea levels and increase the frequency of extreme weather events, the economic risks associated with flooding are expected to intensify.

In response to these recurring tidal threats, the Italian government implemented the MOSE

("*Modulo Sperimentale Elettromeccanico*") barrier system, which was activated for the first time in October 2020 (Silvestri 2020). MOSE is a massive hydraulic infrastructure project with 78 mobile gates installed at the three inlets connecting the Venetian Lagoon to the Adriatic Sea (Bastianello and Balmer 2019). The gates are raised during high tide events when water levels are forecasted to exceed 110 cm, effectively sealing off the lagoon to prevent flooding in the city (Bastianello and Balmer 2019). With an estimated cost of about €5 billion, MOSE is one of Italy's most ambitious climate adaptation projects (Horowitz and Bubola 2023). While MOSE has already demonstrated its capacity to shield Venice from major flood events, its implementation has raised key questions about cost-effectiveness, especially considering delays, maintenance costs, and environmental concerns (Silvestri 2020).

II. Research Question

Prior studies have examined the economic costs of flooding, including infrastructure damage, insurance payouts, and residential property losses (Hallegatte et al. 2013). Specifically in the case of Venice, some papers have analyzed the effect of emergency flood alerts on hotel prices (Angelini et al. 2023), while others have examined MOSE's impact on port activity and flood mitigation (Giupponi et al. 2024). However, relatively little attention has been paid to how tourists respond behaviorally to recurring environmental shocks like "*acqua alta*". This paper fills that gap by offering empirical evidence on how tidal flooding depresses tourism demand in a historic urban center and evaluating whether the preservation of this economic activity alone can justify the cost of public adaptation measures like MOSE. The findings are especially relevant for policymakers weighing the cost-effectiveness of large-scale infrastructure projects aimed at protecting key economic sectors from climate-related disruptions.

We argue that tidal flooding in Venice creates an environmental friction that distorts demand in the tourism sector, leading to a market failure: tourism is depressed below its efficient level due to uncertainty, inconvenience, and reduced access. Because individual tourists and businesses cannot internalize or offset this risk, valuable economic activity is lost that cannot be corrected by private actors alone. Public investment in adaptive infrastructure like the MOSE barrier system can correct this distortion by mitigating flooding, restoring efficient tourism flows, and reducing long-term economic losses. This paper, therefore, asks:

To what extent can the economic costs of flooding, particularly in the form of reduced tourism, justify the €5 billion investment in MOSE? In other words, can tourism alone provide sufficient economic rationale for such a large-scale intervention? We hypothesize that tidal flooding leads to a measurable decrease in tourist arrivals, and that the present value of those cumulative losses over time is comparable to the cost of MOSE.

To address our research question, we examine historical data on flood frequency and tourist arrivals in Venice between 2003 and 2018. We focus on understanding the relationship between monthly flood events and fluctuations in tourism, using this as a basis to estimate the broader economic implications of climate inaction. By focusing on tourism, a quantifiable and policy-relevant economic sector, we aim to contribute to the growing body of literature assessing the benefits of climate adaptation infrastructure in vulnerable urban centers.

III. Data Construction & Evaluation

This study draws on publicly available data from the City of Venice and the Veneto Region Statistical Office to examine the relationship between flooding and tourism in Venice. Meaningful data construction efforts were required to convert disparate datasets into a monthly panel that could support regression analysis.

Flood events were defined as any tide exceeding 110 centimeters, the official threshold at which flooding begins in Venice (Bastianello and Balmer 2019). Hourly tide measurements were obtained from the City of Venice's historical archive of sea level values, which records the exact tide height in centimeters at each hour of each day since the early 2000s (City of Venice 2024). From this dataset, we manually identified all flooding events over 16 years (2003-2018), counting the number of distinct days per month when the tide exceeded 110 cm. This allowed us to construct a monthly time series of flood occurrences, enabling direct comparison to monthly tourism statistics.

Monthly tourist arrivals were taken from the Veneto Region Statistical Database, which provides monthly counts of overnight stays by both domestic and international tourists in the municipality of Venice (Veneto Region 2024). These data were in separate accounts and needed to be aggregated into a singular dataset for total tourist arrivals per month. Annual tourism figures were also obtained from Statista to validate the broader trend and check for consistency (Statista 2025). While slight discrepancies exist between the monthly and annual totals, possibly due to rounding, reporting lags, or differences in classification, this serves as a useful robustness check. The general consistency of patterns across both datasets increases confidence in the accuracy of the monthly data used for regression analysis.

One of the key strengths of the dataset is its temporal granularity. Both floods and tourist arrivals were recorded at high frequency, hourly in the case of tides and monthly for tourism, which allows for month-by-month variation to be exploited in the analysis. This makes it possible to identify short-term behavioral responses to environmental shocks, such as tourists altering their travel decisions in response to flooding events in each month. Moreover, by working with monthly data across 16 years, the analysis naturally controls for recurring seasonal patterns, such as higher tourist volumes in the summer months, helping to isolate the impact of floods from predictable fluctuations. Another advantage is the official nature of the data sources. Both the tide levels and tourism flows come from government statistical offices, which improves reliability and transparency. Additionally, having access to both monthly and annual aggregates allow for cross-validation of trends and strengthens the internal consistency of the analysis.

Despite its strengths, the dataset has several limitations. First, the tourism data only accounts for overnight visitors and excludes "day-trippers", a significant group of tourists in Venice, where same-day visits are common due to cruise ship traffic and regional travel, given the difficulty of accessing this data. Since day-trippers are more likely to alter plans in response to bad weather or flooding, the dataset may underestimate the total impact of floods on tourism. Second, the flood variable may overstate flood exposure due to how events were counted. Rather than grouping multiple tidal surges within a single day as one event, each exceedance of the 110 cm threshold was counted individually. While this captures the intensity of flooding, it may inflate the perceived frequency from a tourist's perspective. Third, the data do not account for the duration of each flood's disruption – how long areas of the city remained inaccessible or how quickly services resumed, which likely affects tourist behavior but is not visible in monthly aggregates. These limitations suggest that our estimates, while informative, may be conservative or miss subtler behavioral responses to flood-related uncertainty.

Nevertheless, the constructed monthly panel forms a solid empirical foundation for identifying general patterns between environmental disruptions and tourism flows in Venice. The temporal granularity and consistency of trends lend confidence to our identification strategy. Ultimately, this dataset allows us to uncover meaningful relationships that are critical for evaluating the economic justification for climate adaptation investments like MOSE.

IV. Data Analysis

To investigate the economic impact of flooding on tourism in Venice, we analyzed monthly data on tourist arrivals and flood frequency from January 2003 to December 2018. Our goal was to estimate the extent to which flooding events, defined as tides reaching or exceeding 110cm, reduce tourist activity, and to assess whether the cost of the MOSE flood barrier is economically justified by the protection it offers to the tourism sector alone.

We first constructed a dataset in which each observation represents one month. The key variables were: (i) the number of floods in that month, and (ii) the difference between observed tourist arrivals and a counterfactual estimate based on constant exponential growth in arrivals from the beginning of the period. This accounts for the long-run upward trend in tourism over time and allows us to isolate deviations potentially caused by flooding, rather than secular growth in global travel demand.

Figure 1 presents a scatter plot of this relationship, with the number of floods per month on the x-axis and the tourist arrival deviation on the y-axis:

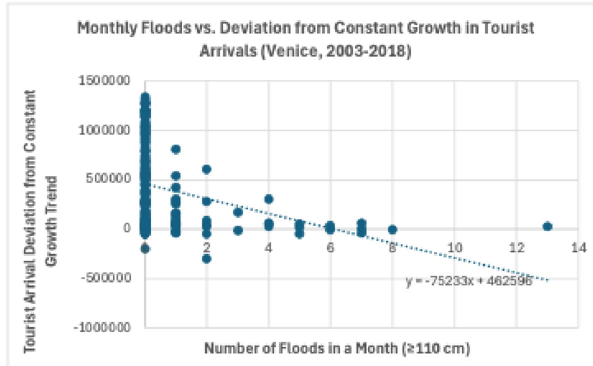

Figure 1. *Monthly Floods vs. Deviation from Constant Growth in Tourist Arrivals (Venice, 2003-2018).*

Note: Monthly data from 2003-2018. Floods are defined as tides ≥ 110cm.

The trend line suggests a negative linear relationship, confirmed by a correlation coefficient of -0.3. A simple linear regression yields the following estimated equation:

(1)

$$TouristLoss_t = \beta_0 + \beta_1 \cdot Floods_t + \varepsilon_t$$

$$TouristLoss_t = 462{,}596 - 75{,}233 \cdot Floods_t$$

This result implies that each additional flood in each month is associated with a decline of approximately 75,233 tourist arrivals, relative to what would be expected under normal growth. We interpret this as evidence that high tide flooding materially depresses tourism in Venice, at least in the short run.

To quantify the economic implications of this tourism loss, we calculate the average loss in monthly revenue associated with one additional flood. The formula is as follows:

(2)

$$MonthlyLoss_t = \Delta Tourists \cdot SpendPerDay \cdot AvgNights$$

Where, $\Delta Tourists$ is 75,233 from our regression result. $SpendPerDay$ is €140, drawn from Valcárcel (2018), and $AvgNights$ is 2.28875, based on the City of Venice tourism data from 2011-2018. Substituting in:

$$MonthlyLoss_t = 75{,}233 \cdot 140 \cdot 2.28875 = €24{,}106{,}534.03$$

Using our flood dataset, we calculate that Venice experienced an average of 7.6875 floods per year between 2003 and 2018. Multiplying this by the estimated loss per flood gives:

$$AnnualLoss = 7.6875 \cdot €24{,}106{,}534.03 = €185{,}318{,}980.32$$

This represents a conservative estimate of the annual tourism revenue loss attributable to flooding, as it only accounts for short-term reductions in tourist arrivals and does not incorporate broader spillover effects on reputation, booking behavior, or cancellations.

To assess whether the MOSE system, which was built to prevent such flooding, is justified from a cost-benefit perspective, we calculate the present value (PV) of these avoided losses over a 50-year horizon. This time frame reflects the MOSE barrier system's intended operational lifespan, as it was designed and implemented with the expectation that it would protect Venice for approximately 50 years. Using a 3% discount rate, consistent with European public infrastructure evaluation standards, we apply the annuity formula:

(3)

$$PV = 185{,}318{,}980.32 \cdot \left(\frac{1-(1+0.03)^{-50}}{0.03}\right) = €4{,}768{,}213{,}629.58$$

Thus, the present value of cumulative avoided tourism losses is approximately €4.77 billion, a figure strikingly close to the estimated cost of constructing the MOSE system, reported at around €5 billion.

It is worth emphasizing that this analysis considers only tourism-related economic benefits and does not include broader effects such as protection of cultural heritage, residential and commercial property, public infrastructure, or local livelihoods. When these are considered, the full value of the MOSE system would likely exceed the estimate derived here.

V. Policy Implications

The results of this analysis carry several important implications for climate adaptation policy in Venice and similarly vulnerable cities. Most directly, the finding that each flood event leads to a measurable and economically significant reduction in tourist arrivals strengthens the case for public intervention to stabilize tourism demand. In a context where private actors, hotels, restaurants, and visitors cannot predict or mitigate flooding risks on their own, a large-scale public adaptation investment like the MOSE barrier system functions as a corrective mechanism that restores market efficiency.

Importantly, this framework repositions climate adaptation not merely as a matter of disaster prevention or heritage preservation, but as a tool for safeguarding long-term economic activity. By reducing uncertainty and maintaining access to the city during high-tide events, MOSE lowers the friction that would otherwise depress tourism below its efficient level. This demand stabilization is particularly valuable in Venice, where tourism represents a disproportionately large share of local income, employment, and municipal revenue.

That said, the economic case for MOSE should be evaluated considering both its strengths and limitations. While the present discounted value of avoided tourism losses comes close to the estimated €5 billion construction cost, this benefit estimate is based on a narrow category. It excludes less tangible but important spillovers, such as reputational damage, long-term shifts in traveler preferences, and the broader effects on property markets. Nor does it factor in MOSE's maintenance costs, potential environmental impacts on the lagoon ecosystem, or the institutional and political challenges that have affected its rollout.

Therefore, policymakers should view MOSE not as a one-time fix, but as part of a broader, long-term strategy for climate resilience. This should include complementary measures such as investments in sustainable tourism practices, resilient transportation and drainage infrastructure, early-warning systems, and community engagement initiatives that build local preparedness. Moreover, as sea levels continue to rise, the thresholds at which MOSE is triggered may need to be reassessed to balance flood protection with ecological sustainability.

Finally, the findings suggest that similar cost-benefit reasoning could apply to other historic cities exposed to sea-level rise, especially those with tourism-dependent economies. Cities like Amsterdam, Bangkok, and coastal areas of Japan may benefit from conducting comparable analyses to assess whether adaptive infrastructure can mitigate market failures linked to climate risk and help stabilize critical economic sectors.

VI. Conclusion

This paper examined the relationship between tidal flooding and tourism in Venice, using monthly data from 2003 to 2018. We found that each additional high-tide flood is associated with a decline of approximately 75,233 tourist arrivals relative to a constant growth baseline. This relationship is reflected in a negative correlation of -0.3 and confirmed by linear regression analysis. When translated into economic terms, the estimated annual loss in tourism revenue from flooding is substantial, and when projected over a 50-year horizon and discounted at 3%, the present value of these cumulative losses approaches €4.77 billion.

These findings offer a compelling case for the economic justification of flood prevention infrastructure in a tourism-dependent city like Venice. By treating flooding as an environmental friction that depresses tourism below its efficient level, we frame the MOSE barrier system as a policy response that corrects a persistent market failure. Public investment in flood protection not only safeguards physical assets and cultural heritage but also plays a critical role in preserving long-term economic stability. As sea levels continue to rise and extreme weather events become more frequent, the value of forward-looking adaptation policies becomes not only a question of resilience, but of economic necessity.

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